

Smart Tips to Design a Hybrid/Electric Vehicle

Wensi Jin, automotive industry manager, MathWorks shares his thoughts on the top three design considerations that engineers should keep in mind while designing hybrids and EVs. He has wide experience in this space as MathWorks' solutions are used by industry majors such as Tesla, GM, ATB Technologies and Dongfeng Motor Corporation.

Carry out design optimisation from the vehicle level and down.

After decades of refinement, development of conventional vehicles tends to follow an evolutionary path. Making incremental improvements is the norm. Instead of new powertrain architecture, the focus is often to improve engine, transmission and other subsystems. In contrast, with hybrids and EVs comes a wide range of powertrain architecture options. The design space to be explored becomes much vaster. Once an architecture is selected (for example, range-extended EV with a gasoline engine as the generator), each subsystem needs to be sized based on the often-competing requirements such as performance, cost, weight and thermo. It is through this design process that hundreds of possible design choices are narrowed down to a few candidate designs, based on which prototype vehicles are developed. As it is impossible to test every design choice using prototype vehicles, simulation is deployed during architecture evaluation, subsystem sizing and other trade-off studies. Naturally, these simulations need to be carried out on the vehicle level, following predefined drive cycles. In other words, each vehicle model is 'driven' through a series of scenarios for the purpose of assessing their performance and other desired characters. The drive cycles or scenarios are anywhere from minutes to tens of minutes. Therefore design engineers need to create a proper balance in the level of details in the model in order to gain simulation speed. These simulations are usually run between near real-time and several times faster than real-time. Fast simulation speed allows design engineers to run hundreds of iterations for each design choice. At the same time, design engineers need to choose simulation tools that allow them to

build up a vehicle level simulation model quickly, due to the number of design choices they have to evaluate.

Master the battery, motor and energy management technologies.

Hybrids and EVs are enabled by a number of subsystems that are not used in conventional vehicles, including advanced batteries as the main energy storage device, electric motors which provide power assist or propel the vehicle and an energy management system. There are many implications of adding these new technologies. First of all, automotive companies need to choose which technologies to develop in-house from ground up and which technologies to co-develop with partner companies. Second, regardless of the source of the new technologies, engineering talent needs to be in place to master these technologies so the new subsystems can be efficiently specified, designed and verified. For example, many automotive companies did not have in-house specialists in advanced batteries until recently. Now, these companies have several specialised teams for battery development. Third, with the new technologies and new engineers, development processes and tool chains need to be carefully defined and rolled out. This last point is particularly important on the engineering level. Due to the design complexity in advanced battery, electric motor and energy management systems, model-based design has become the de facto development method for these subsystems, including the embedded control software inside.

Gain competency in embedded control software. Virtually every new subsystem for hybrids and EVs is controlled by embedded software. Unlike conventional vehicles, whose many subsystems are mechanically oriented, hybrids and EVs contain significantly higher electronics and embedded software content, which can add up to 35 per cent or more of the cost of the vehicle. Automotive companies that develop hybrids and EVs must develop internal competencies in embedded software, including the management of embedded software development and quality.

GM's OnStar and Hyundai's BlueLink are much appreciated remote diagnostic tools. Toolmakers such as Delphi and LandAirSea Systems provide plug-in remote diagnostic solutions, which connect to your car and to a cloud (through a wireless network) to provide a host of online diagnostic solutions to any car produced post 1996.

More updates.

► The number of ECUs in a car also means that these modern vehicles are at risk from hackers! In order to allay the fears of car users, Harman International Industries is busy developing a software security system that will protect at least the vital ECUs, such as the engine management system, from hackers, if not the infotainment aspects. The solution is expected to be in cars in the 2016-17 timeframe.

► Bosch recently demonstrated advanced MEMS sensors for applications such as skid detection, adaptive cruise control (ACC) and softer gear changing of automatic gearboxes. According to a presentation at CES, a modern car has at least 50 such MEMS sensors, which in turn are supplying data to the automotive cloud that supports drive assistance systems and telematics.

My vehicle, my conscience

The number of vehicles in the world is growing far too fast, leading not just to a depletion of oil reserves but scary levels of pollution as well. Car makers, with a due share of eco-responsibility, are introducing several energy management technologies in all new cars, as well as launching electric or hybrid models for a cleaner and sustainable future.

"In the last few years, there has been a significant focus towards advanced engine control systems to offer higher fuel efficiency (including hybrid technology) and near-zero emissions. The start-stop technology, which is still in a nascent stage, is one step in this direction," says Gupta. "Another interesting thing that needs to be done to ensure we get greater mileage and lower emissions than we are capable of today, is to make the cars lighter in weight. Experts suggest that one solution is to make body components of lighter materials such as carbon fibre-reinforced plastic (CFRP) or polymer. CFRP works much like fibreglass; the carbon fibre is spun into long strands and then arranged in a cloth-like weave for strength. The Z06

version of the Chevrolet Corvette already makes limited use of carbon fibre right from the factory."

Electric and hybrid vehicles are becoming quite popular, with Ford, Honda, Nissan, Toyota, Mahindra Reva and Renault being significant players in the market here. "Availability of higher capacity, 48-volt batteries, adoption of fuel cell vehicles and natural gas cutting into the electric truck market are some notable trends in this space," says Gupta.

Maini of Mahindra Reva points out that apart from the light-weighting efforts explained by Gupta earlier, another key development in the electric car space is related to battery technology. "A suitable way of storing energy with the space constraints in mind was the biggest challenge. If the weight of the battery is really high, most of the power generated will be used to carry the car's battery alone, hence considerably affecting the car's performance. However, the growth in the smartphone industry has caused battery technology to hit its peak on the learning curve. A lithium-ion battery pack can store energy for moving the car and at the same time be carried around on the car, tipping the balance at only 830kg for the entire